

Control Principles of Complex Networks

An Implementation on Contact Matrices in Protein Folding

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Abstract—For complex networks, possessing non-trivial topologies and often evolving through a set of non-linear dynamical equations, the control problem poses challenges that classical control theory cannot always answer. In the last two decades, studying control principles of such systems has attracted much attention. So far, many seminal works and fundamental theories have been presented in this field, and much progress has been achieved. Here, we will go through some of the fundamental theories regarding both linear and non-linear control systems. In the end, we will utilize two well-known linear and non-linear control frameworks to evaluate the controllability of the circuit topology networks associated with protein folding and identify the driver nodes in both frameworks.

I. LINEAR CONTROLLABILITY

In the well-known work, Kalman states that a linear system is *controllable* if we can drive it from any initial state to any desired final state in finite time [1]. Recent advances in the controllability of complex networks offer mathematical tools to identify the driver nodes: a subset of nodes whose direct control with appropriate signals can control the state of the full system.

A. Linear Time-Invariant Systems

For a large scale analysis, a common starting point in the literature on the control of complex networks is the linear time-invariant (LTI) control system and a typical dynamics for which may be written as:

$$\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t) + \mathbf{B}\mathbf{u}(t). \quad (1)$$

Consider the LTI dynamics, on a network $G(\mathbf{A})$ of N nodes. The N nodes of the graph are associated with a vector of finite size consisting of state variables $x_i(t)$, such that $\mathbf{x}(t) \in \mathbf{R}^n$. The state variable $x_i(t)$ can, for example, denote the amount of traffic that passes through the node i on a communication network or voltage fluctuations in measured EEG time series in given sensor.

The state matrix $\mathbf{A} := (a_{ij})_{N \times N}$ represents the weighted wiring diagram of the underlying network, where a_{ij} is the strength or weight with which node j affects/influences node i . The elements of the adjacency matrix can be interpreted differently in different contexts e.g., in neuroscience, a positive (or negative) a_{ij} means the link ($j \rightarrow i$) is excitatory (or inhibitory), and $a_{ij} = 0$ if node j has no direct influence on node i , typically.

Consider M independent control signals $\{u_1, \dots, u_M\}$ applied to the network. The input matrix $\mathbf{B} := (b_{im})_{N \times M}$ identifies the nodes that are directly controlled, where b_{im}

represents the strength of an external control signal $u_m(t)$ injected into node i .

The input signal $\mathbf{u}(t) = (u_1(t), \dots, u_M(t))^T \in \mathbf{R}^M$ can be imposed on all nodes or only a preselected subset of the nodes. In general, the same signal $u_m(t)$ can drive multiple nodes. The nodes directly controlled by $\mathbf{u}(t)$ are called *actuator nodes* or simply *actuators*, like nodes x_1 , x_2 and x_5 in Fig. 1. The number of actuators is given by the number of non-zero elements in $\mathbf{B}$. The actuators that do not share input signals, e.g. nodes x_1 and x_2 in Fig. 1, are called *driver nodes* or simply *drivers*.

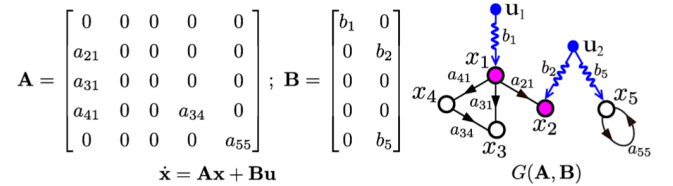


Fig. 1: Representation of an LTI system under control.

B. The Kalman's Criterion of Controllability

The best-known controllability tests is Kalman's rank condition stating that the LTI system $(\mathbf{A}, \mathbf{B})$ is controllable if and only if the controllability matrix:

$$\mathbf{C} = [\mathbf{B}, \mathbf{A}\mathbf{B}, \mathbf{A}^2\mathbf{B}, \dots, \mathbf{A}^{N-1}\mathbf{B}] \quad (2)$$

has full rank, i.e.

$$\text{rank } \mathbf{C} = N. \quad (3)$$

If, however, $\text{rank } \mathbf{C} < N$, then we can find an appropriate input vector $\mathbf{u}(t)$ to steer the system from $\mathbf{x}(0)$ to an arbitrary $\mathbf{x}(t)$. Hence, the system is controllable.

Example: In the network control problem of Fig. 2a the controllability matrix:

$$\mathbf{C} = \begin{bmatrix} b_1 & 0 & 0 \\ 0 & a_{21}b_1 & 0 \\ 0 & a_{31}b_1 & 0 \end{bmatrix} \quad (4)$$

is always rank deficient, as long as the parameters b_1 , a_{21} and a_{31} are non-zero. Hence, the system is uncontrollable. In contrast, for Fig. 2c we have:

$$\mathbf{C} = \begin{bmatrix} b_1 & 0 & 0 & 0 & 0 & 0 \\ 0 & b_2 & a_{21}b_1 & 0 & 0 & 0 \\ 0 & 0 & a_{31}b_1 & 0 & 0 & 0 \end{bmatrix} \quad (5)$$

which has full rank, as long as the parameters b_1 , b_2 , a_{21} and a_{31} are non-zero. Hence the system is controllable.

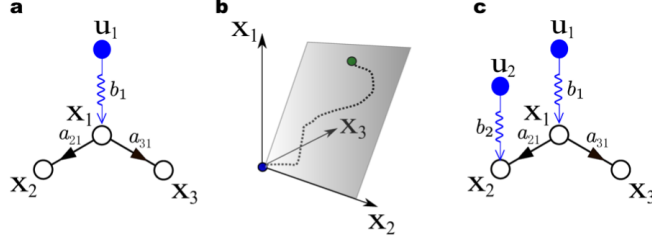


Fig. 2: Examples of networks attaining full controllability(c) / not attaining full controllability(a), stuck on a 2-d phase plane(b).

C. Structural Controllability

For many complex networks, a_{ij} (e.g. the elements in A) are not precisely known. Indeed, we are often unable to measure the weights of the links, knowing only whether there is a link or not. Hence, if not conceptually impossible, it is hard to verify Kalman's rank condition using fixed weights numerically. Structural control, introduced by Lin in the 1970s, offers a framework to avoid this limitation systematically [9].

An LTI system (A, B) is a structured system if the elements in A and B are either fixed zeros or independent free parameters. Hence, A and B are called structured matrices. The system (A, B) is structurally controllable if we set the nonzero elements in A and B . The resulting system is controllable in the usual sense (i.e., $\text{rank } C = N$). This leads to the construction of the state matrix to show the absence of the link with zero and its existence with only one.

D. Minimum Inputs Theorem

In the significant and much noticed work of Liu [9], it has been proven that to fully control a network $G(A)$ with the minimum number of input vertices (or equivalently the minimum number of driver nodes), we need the size of the maximum matching in the corresponding digraph $G(A)$. Here, we introduce the concept of matching in a directed/undirected graph.

Definition: For an undirected graph, a matching M is an independent edge set, i.e. a set of edges without common vertices. A vertex is matched if it is incident to an edge in the matching. Otherwise, the vertex is unmatched. In a directed graph, the condition on the edges would be different such that the edges without common starting or ending points belong to the set of matching, and the vertex they are pointing to would be matched.

A matching of maximum cardinality/size is called a maximum matching. We note that, in general, there could be many different maximum matchings for a given graph or digraph. A maximum matching is called perfect if all vertices are matched. For example, in a directed elementary cycle,

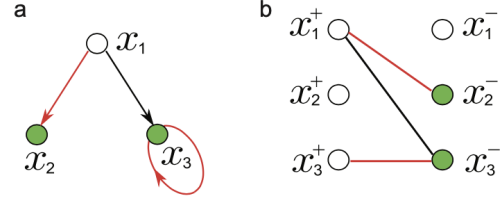


Fig. 3: Maximum matching in a digraph(a) and its undirected bipartite map(b).

all vertices are matched. Note that a maximum matching of a digraph $G(A)$ can be easily found in its bipartite representation, denoted as $H(A)$, see Fig. 3. The bipartite graph is defined in the following way, $H(A) = (V^+ \cup V^-, \Gamma)$, where $V_A^+ = \{x_1^+, \dots, x_N^+\}$ and $V^- = \{x_1^-, \dots, x_N^-\}$ are the set of vertices corresponding to the N columns and rows of the state matrix A , respectively. Edge set will be $\Gamma = \{(x^+, x^-) | a \neq 0\}$. For a general bipartite graph, its maximum matching can be found efficiently using the well-known Hopcroft-Karp algorithm, which runs in $O(\sqrt{VE})$ time.

E. Control Energy

Identifying the minimum number of driver or actuator nodes sufficient for control is only the first step of the control problem.

Consider the LTI system (1) driven from an arbitrary initial state $\mathbf{x}_i$ towards a desired final state $\mathbf{x}_f$ by the external signal $\mathbf{u}(t)$ in the time interval $t \in [0, T]$. We define the associated control effort in the quadratic form as:

$$\mathcal{E} \equiv \int_0^{t_f} \|\mathbf{u}(t)\|^2 dt, \quad (6)$$

called the "control energy" in the literature.

For a fixed set of driver nodes, the control input $\mathbf{u}(t)$ that can drive the system from $\mathbf{x}_i$ to $\mathbf{x}_f$ can be chosen in many different ways, resulting in different trajectories followed by the system. Each of these trajectories has its control energy. Of all the possible inputs, the one that yields the minimum control energy is

$$\mathbf{u}(t) = \mathbf{B}^T e^{\mathbf{A}^T(t_f-t)} \mathbf{W}_B^{-1}(t_f) \mathbf{x}_f \quad (7)$$

where $\mathbf{W}_B(t)$ is the gramian matrix and $\mathbf{x}_f = e^{\mathbf{A}t_f} \mathbf{x}_0$.

$$\mathbf{W}_B(t) \equiv \int_0^{t_f} \exp(\mathbf{A}\tau) \mathbf{B} \mathbf{B}^T \exp(\mathbf{A}^T \tau) d\tau \quad (8)$$

Without loss of generality, we can set the final state at the origin, $\mathbf{x}_f = 0$, and write the control energy as

$$\mathcal{E} = \mathbf{x}_0^T \mathbf{H}^{-1}(t_f) \mathbf{x}_0 \quad (9)$$

where $\mathbf{H}(t_f) = e^{-\mathbf{A}t_f} \mathbf{W}_B e^{-\mathbf{A}^T t_f}$ is the symmetric Gramian matrix.

Using the Rayleigh-Ritz variational method, the normalized control energy obeys the bounds

$$\eta_{\max}^{-1} \equiv E_{\min} \leq E(t_f) \leq E_{\max} \equiv \eta_{\min}^{-1} \quad (10)$$

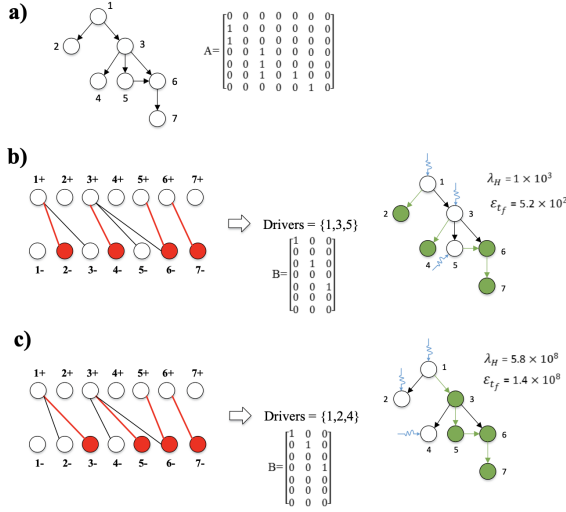


Fig. 4: Maximum Matching. Control of the network in figure (a), there are two possible set for diver nodes, however one of them has minimum control energy, set (b).

where η_{max} and η_{min} the maximum and minimum eigenvalues of $\mathbf{H}$, respectively.

We consider a simple model: a unidirectional, one-dimensional string network, for which an analytic estimate of the control energy can be obtained as

$$\mathcal{E}_l \approx \lambda_H^{-1} \quad (11)$$

where l is the chain length (the number of nodes on the link) and λ_H is the smallest eigenvalue of the symmetric Gramian matrix.

The matrix H is positive definite and symmetric with its inverse satisfying $H^{-1} = Q\Lambda Q^T$, where Λ and Q are the corresponding eigenvalue and eigenvector matrices, respectively. Thus, λ_H^{-1} is the largest eigenvalue of H^{-1} with λ_H denoting the smallest eigenvalue of H . Also $\mathbf{q}_i$ is the i th column of Q . If the initial state x_0 is chosen to satisfy $\mathbf{q}_1^T \cdot x_0 = 1$, we obtain $\mathcal{E}(T) \approx \lambda_H^{-1}$.

F. Concepts of longest control chains

Identifying maximum matching so that the network is deemed structurally controllable does not guarantee convergent control energy.

To find the longest control chain (LCC) in a network, we first use the maximum matching algorithm to find all the driver nodes. We then identify the shortest paths from each of the driver nodes to each of the non-driver nodes. Finally, we pick out the longest such shortest paths as LCCs. The computational complexity of finding the LCCs is associated with the maximum matching algorithm plus searching for the shortest paths, which is feasible for large networks. There can be multiple LCCs. The node at the end of an LCC is most challenging to control because the most considerable amount of control energy is required. The number m of such end nodes dictates the degeneracy (multiplicity) of LCCs.

An example is shown in Fig. 4., where we see that, there are multiple LCCs.

In Fig. 4b, Nodes 1, 3 and 5 are driver nodes. For this network $LCC = 2$ and in Fig. 4c driver nodes are 1,2 and 4. For the set of driver nodes we considered in part c, the longest control path is equal to three, $LCC = 3$. As it turns out, any increase in LCC is accompanied by a huge increase in λ_H^{-1} . So we can say that LCC is an important parameter in determining the control energy.

G. Projected Gradient Method

By fixing the quadratic sum of all elements of matrix B and substituting (7) into (6), the minimum-cost control problem can be rewritten as:

$$\min \quad \text{tr}[W_B^{-1} X_f] \quad (12)$$

s.t.

$$\text{tr}(B^T B) - M - \varepsilon = 0 \quad (13)$$

where X_f is a constant matrix given by $X_f = e^{A t_f} X_0 e^{A^T t_f}$.

$E(B) \triangleq \text{tr}[W_B^{-1} X_f]$ is the defined energy cost function to be minimized. We also define a norm function $N(B) \triangleq (\text{tr}(B^T B) - M)^2$ to associate with the equality constraint expressing the normalization condition on B , where ε is a positive constant to ensure that $N(B)$ is non-zero on the surface $\text{tr}(B^T B) = M + \varepsilon$.

The projected gradient method (PGM) is an efficient algorithm to solve the optimization problem defined above. The main idea is to analytically derive the derivatives of the functions $N(B)$ and $E(B)$ with respect to B , and then project the negative of the gradient of $E(B)$ onto the sphere surface $\text{tr}(B^T B) = M + \varepsilon$. By doing so, B can be searched via an iterative process until convergence. The gradients of functions $N(B)$ and $E(B)$ can be obtained as:

$$\frac{\partial N(B)}{\partial B} = 4[\text{tr}(B^T B) - M] \cdot B \quad (14)$$

$$\frac{\partial E(B)}{\partial B} = - \int_0^{t_f} e^{A^T t} W_B^{-1} X_f W_B^{-1} e^{A t} dt \cdot B \quad (15)$$

respectively. The two gradients point to the directions where $N(B)$ and $E(B)$ have the fastest increasing speed respectively. We also show that $\partial E(B)/\partial B$ is always non-zero as long as (A, B) is controllable, which explains the importance of imposing the normalization condition on B . In the convergence the PGM algorithm provide us minimum-cost control driver nodes. PGM tries to find optimal set of driver nodes regarding the amount of needed energy. Like any other gradient descent algorithms, PGM also works in an iterative manner. As the result, to find the global minimum energy, number of iteration has to tuned.

II. NON-LINEAR CONTROLLABILITY: USING ONLY THE STRUCTURE OF UNDERLYING NETWORK

So far, we have reviewed some of the essential concepts of linear controllability. Linear dynamics is the most convenient and rather comfortable primary approach in many problems for that it can bypass many unnecessary complexities. However, in the real world, most systems evolve through a set of non-linear dynamical equations. Also, linear simplifications will be reliable only in the vicinity of a dynamical fixed point or a nominal trajectory and can become misleading in predicting non-local behaviors. These issues started a line of studies focused on the controllability of non-linear dynamical systems.

At first, researchers were looking for a set of criteria like that of Kalman's criterion, which will guarantee the controllability as defined previously -ability to explore the whole phase space in finite time through appropriate inputs- for any generally non-linear system. However, these attempts soon proved to be too ambitious to work, in general. Also, since most real systems manifest their behavior in few stable states, the attention was soon drawn towards stabilizing the current state through control inputs or generally controlling the system's transition towards the desired final state.

In the following, we will review a well-known and relevant work, known as "feedback vertex set control", which is mainly developed in the biological context and considers general non-linear dynamics with some easily satisfiable conditions. This method uses mostly the structure of underlying network.

A. Feedback Vertex Set Control

As an approach to control (steer) a general non-linear system with subsystems wired through arbitrary non-trivial network topology towards a final state which should belong to the systems' dynamical invariant set, "feedback vertex set control" was introduced in [5]. This approach is based on combining concepts "determining nodes" from dynamical systems and "feedback vertex set" from graph theory.

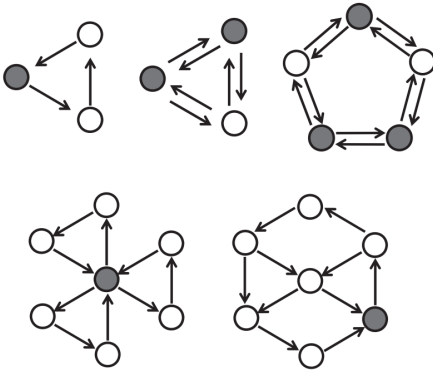


Fig. 5: Examples of Feedback Vertex Set (gray nodes).

Feedback vertex set are in fact a set of nodes in a digraph such that their removal will leave the graph with no loops Fig. 5.

Fig. 5. Also, determining elements (subsystems/nodes) are referred to as those subsystems for which the convergence of their variables for two arbitrary paths guarantees the convergence of the whole system. In other words, if x_1 and x_2 are two different solutions for the system, they are guaranteed to converge in the limit of large times if the determining nodes converge. As a result, if the dynamics of the determining nodes are given for large times, then the dynamics of the whole system is determined uniquely for large times. In their work, they have proved mathematically that [5]:

Any feedback vertex set of a regulatory network determines nodes of the dynamics on the network. Conversely, if a vertex set is determined for all choices of nonlinearities compatible with the network structure, it is a feedback vertex set..

The approaches based on control of nonlinear dynamics, which use only the properties of the underlying network, are applicable for a vast domain of complex systems. The only criteria required to be met is the continuity and dissipativity of the dynamics, which most real world systems meet and satisfy. The exact mathematical foundations of this work make it distinct from many others.

More precisely, in FVS control framework, it has been assumed that the system possesses a general dissipative non-linear dynamics like:

$$\dot{x}_i = F_i(x_i, x_J, t) \quad (16)$$

in which x_J is a regulatory set of nodes (including x_i itself if $\partial F_i / \partial x_i \geq 0$), regulating the state of the node i . Also, the function F and all its derivatives must be continuous while the the volume of the possible states of the system is limited too, i.e. for any finite initial condition at t_0 , we require that for all $t \geq t_0$, $|\tilde{x}(t)| < C$. Now, if in such a system, consisting of N dynamical units, there exists a probably non empty set like $\{J\} \subseteq \{1, 2, \dots, N\}$, such that for dynamical solutions $\tilde{x}^{(1)}$ and $\tilde{x}^{(2)}$, state variables in $\{J\}$ converge i.e.

$$\lim_{t \rightarrow \infty} (\tilde{x}_J^{(1)}(t) - \tilde{x}_J^{(2)}(t)) \rightarrow 0,$$

then for all choices of nonlinearities F , this will be the case:

$$\lim_{t \rightarrow \infty} (\tilde{x}^{(1)}(t) - \tilde{x}^{(2)}(t)) \rightarrow 0,$$

if and only if $\{J\}$ is a feedback vertex set.

We should note that the minimum set of the feedback vertex set will not be unique. Several of such subsets may exist that are minimum in size. This approach falls under the category of open-loop control strategies. The network will be divided into several groups in which one node will lead its sequence in the loop since it is being driven constantly and becomes the leader node in a sense. When driver function is function of time $u(t)$, it is called an open-loop control and if be function of state of system, i.e. $u = u(x)$ it is known as a state feedback (output feedback) closed-loop control.

In the FVS framework, we need to check for continuity, dissipativity, and non-divergent dynamics first. Then, at the next step, we need the network structure on which the dynamics evolve (to look for the FVS set) and the dynamics of the FVS nodes in the final target state to lock them onto it.

B. Minimum Dominating Set Control

One of the other frameworks introduced for assessing the controllability of generally non-linear dynamics on an undirected network is a framework based on a graph-theoretical concept called the dominating set (DS). A DS is a subset of nodes in a graph $G(V, E)$, such that each node in V is either in that subset or just adjacent to it. Now, such a subset with the minimum cardinality is a minimum dominating set (MDS). Examples are illustrated in Fig. 6. One notice that the MDS nodes can possess different configurations as in the maximum matching algorithm. Identifying an MDS is an NP-hard problem; however, it can be mapped onto a linear integer programming problem through which even the large scale networks shall be dealt with in relatively short times.

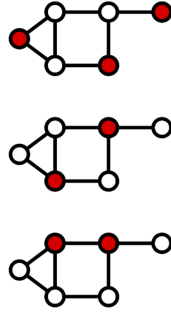


Fig. 6: Different configurations of dominating sets in red color; the middle and the below are MDS.

To relate the concepts of MDS and network controllability, a theorem states that supposing that every edge in a network is *bi-directional* and *every node in the MDS can control all of its outgoing links separately*, then the network is structurally controllable by selecting the nodes in the MDS as driver nodes. It can be proved how this framework can guarantee structural controllability. An illustration of the difference between controllability in its basic Kalman sense and in MDS control sense is given in Fig. 7.

To understand the role of different topological features in determining an MDS set or prioritizing some specific characteristics in the network, one would need all possible MDS configurations to make a choice. However, to this day, there are no effective and practical algorithms that can yield all possible configurations or even the size of such a set. A centrality-corrected MDS approach has been proposed recently, in the context of the biological networks based on the premise of centrality-lethality rule, suggesting that the highly connected proteins in the PPI network are more likely to be essential. A CC-MDS can be implemented only

by a small change in the conditions of the linear integer programming associated with weighted MDS evaluation.

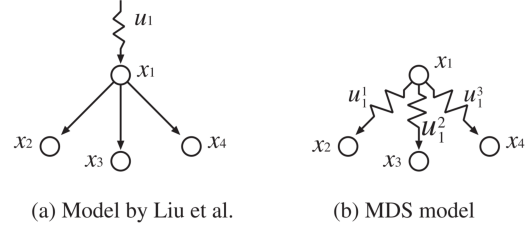


Fig. 7: Concepts of maximum matching control (left) and minimum dominating set control (right).

III. IMPLEMENTATION OF CONTROL THEORIES ON "CIRCUIT TOPOLOGY" NETWORKS

Biological systems reflect some of the most complex manifestations of nature. Understanding them has challenged many generations of scientists, and the multidisciplinary of the research going on in this field has led to many valuable achievements. No doubts that the final goal of all the progress would be to answer the question of how to control their behavior. Here, we will present the results as we have implemented linear and non-linear control strategies to evaluate the controllability of circuit topology networks for protein folding.

At first, we have evaluated the linear controllability endowed with minimum energy constraints. As proposed, we will take the values $S = 0, P = 1, X = 0.5$ for the provided matrices. In Fig. 8, we can see that for control of the networks, one needs a rather small fraction of the nodes (driver nodes are highlighted red), meaning that one can achieve the complete controllability of these systems *-in the linear sense-* through injection of appropriate control inputs to red ones. An interpretation of this could be that the highlighted nodes are the main sites through which the linear stability of the dynamics on such a structure can be guaranteed at the minimum energy. In some sense, maybe targeting some of these control nodes (loops) can reflect in a loss of linear stability the case of environmental perturbations to the protein.

Next, we have also considered the case in which $S = 1, P = 2, X = 3$ and the results are illustrated in Fig. 9. We can see that while the graph has a full structure now, still the driver node fractions are almost the same as before, while some nodes, again, represent themselves as driver nodes as they did in the previous case. In the folder List-of-control-nodes, we provide the number of driver nodes in linear minimum energy control theory.

As another approach, we have implemented the structural part of the non-linear control strategy *-FVS Control-* to get a sense of how many feedback vertex would be needed to make the possible transitions among the dynamical invariant set elements in this system.

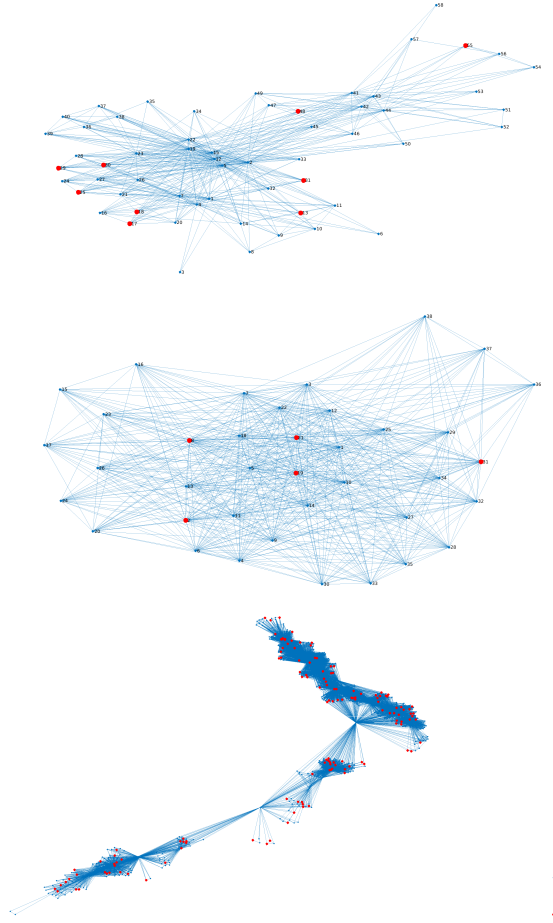


Fig. 8: Linear controllability evaluation of: a6n(upper), aps(center) and chromosome(below) in the regime $S = 0, P = 1, X = 0.5$. The driver nodes are highlighted in red. The associated fraction of driver nodes are 9/58, 5/38 and 138/765, respectively.

Here, we have mapped the undirected graphs onto bidirected ones to use the FVS control framework. We also note that to evaluate the determining nodes (driver nodes) in FVS-C, only the structure of the associated graph matters and nothing more, so, to evaluate FVS-C, we take $S = 0$ and let P, X take any arbitrary non-zero value. The results are illustrated in Fig. 10.

As we can see, the reason for the very high fraction of the FVS is that we have transformed the undirected matrices into bidirected ones, which will make a loop between each pair of connected nodes. Thus, it takes much more nodes to remove all the loops in the graph. However, if we can generate a directed regulatory network under any possible conditions, then applying the FVS control would be much more efficient and much more meaningful. Also, for the case $S = 1, P = 2, X = 3$, all the nodes in the graph would belong to the FVS since the associated structure will be a bidirectional map of a fully connected undirected graph. Between any pair of nodes, there exists a loop.

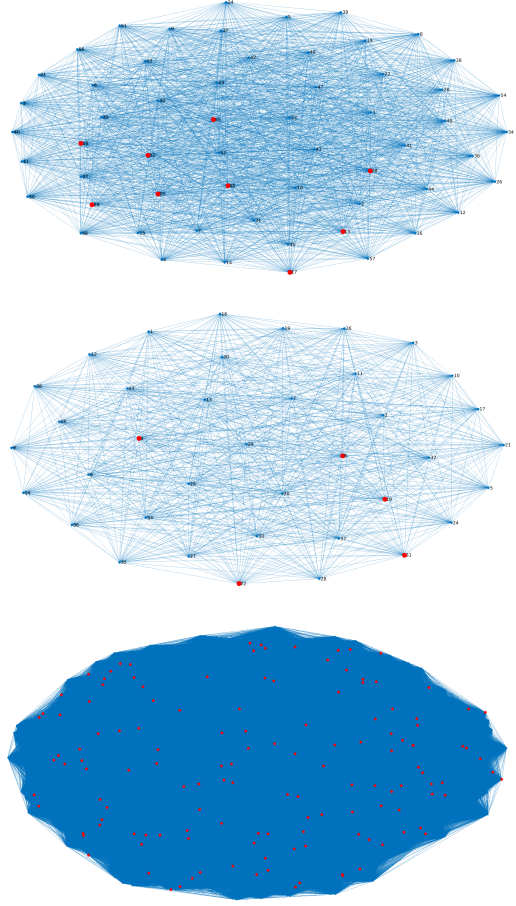


Fig. 9: Linear controllability evaluation of: a6n(upper), aps(center) and chromosome(below) in the regime $S = 1, P = 2, X = 3$. The driver nodes are highlighted in red. The associated fraction of driver nodes are 9/58, 5/38 and 138/765, respectively.

We have also obtained the MDS driver nodes in the same networks used for FVS evaluation, and the results are brought in Fig. 11. Again, we can see how dramatically the size of the required driver nodes is changed (decreased). Still, one should consider other criteria concerning MDS control: each of these nodes needs to control their adjacent nodes independently.

In the folder List-of-control-nodes, we provide the number of driver nodes using FVS control theory.

IV. IMPLEMENTATION OF CONTROL THEORIES ON "CONTACT MAPS" NETWORKS

In recent years, there has been considerable interest in examining the structure and dynamics of complex networks. Proteins in 3D space may also be considered as complex systems that emerge through the interactions of their constituent amino acids.

A way in conceptualizing and modelling protein structures is to consider the contacts between atoms in amino acids as a

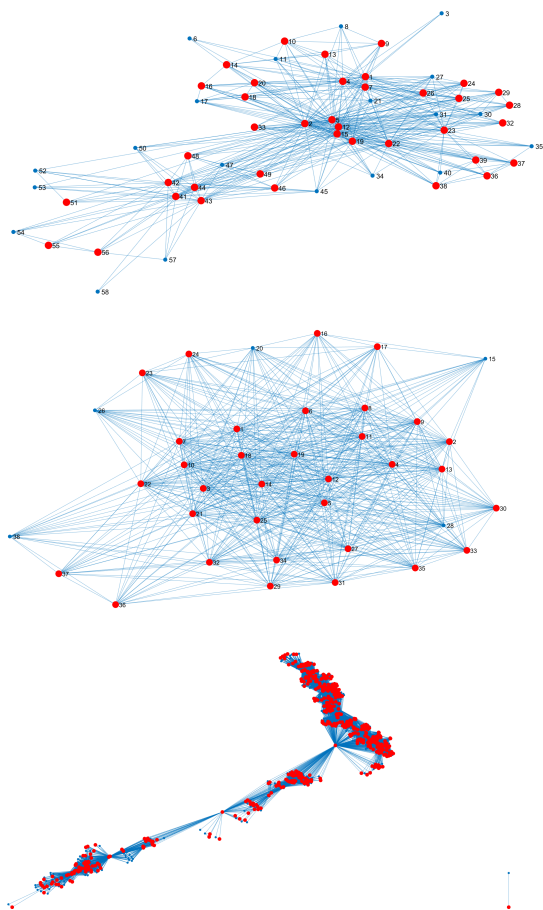


Fig. 10: Non-linear (FVS-C) controllability evaluation of: a6n(upper), aps(center) and chromosome(below) in the regime $S = 0$ and P, X any arbitrary non-zero value. The driver nodes are highlighted in red. The associated fraction of minimum FVS (driver nodes) are 38/58, 33/38 and 522/765, respectively.

network of interactions, irrespective of secondary structures and fold type.

There is a natural distinction of contacts into two types: long-range and short-range interactions. Long-range interactions occur between residues distant from each other in the primary structure but situated at a much closer distance in the tertiary structure. Short-range interactions occur between residues that are local in the primary, secondary and tertiary structures.

For most networks, what is termed as a node and a link is relatively straightforward. When looking at protein transition states, several studies have considered the $C\alpha$ atoms to be the nodes and established a link between two nodes if the atoms were within $8.5 \text{ \AA}$ of each other. Analyses of protein 3D structures as a complex network approach help us understand many different aspects, including its structural flexibility, key residues stabilizing its 3D structures, folding nucleus,

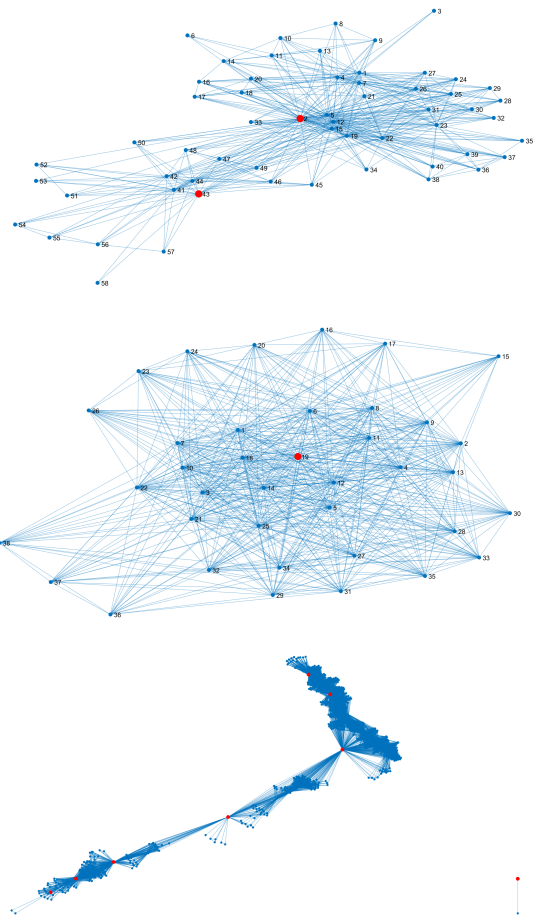


Fig. 11: Non-linear (MDS-C) controllability evaluation of: a6n(upper), aps(center) and chromosome(below) in the regime $S = 0$ and P, X any arbitrary non-zero value. The driver nodes are highlighted in red. The associated fraction of minimum FVS (driver nodes) are 2/58, 1/38 and 8/765, respectively.

important functional residues, mixing behaviour of the amino acids and hierarchy of the structure, etc.

As we can see, there are lots of isolated nodes for contact maps matrices. Therefore, we provide two different graphs and a list of control nodes considering isolation nodes and without.

We have obtained the MDS driver nodes for the contact maps, and the results for the centre cluster of contact maps are brought in Fig. 13.

In the folder List-of-control-nodes, we provide the number of driver nodes using MDS control theory.

V. RESULTS

We provided three tables for different kinds of proteins and methods. In these methods, we used the state-transition matrix.

As shown in Table 1. and Table 2., there is no significant difference between the average of normalized driver node

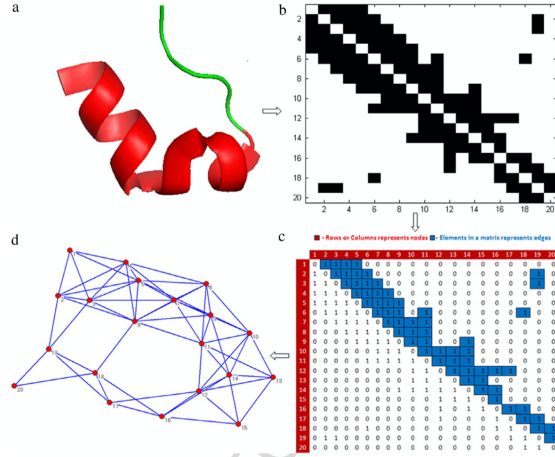


Fig. 12: Modelling of protein structure as a complex network. (a) the alpha helix is depicted in red and coils or loops in green. (b) 2D binary symmetric matrix or adjacency matrix represents protein contact map. (c) Protein contact map representing in adjacency matrix and (d) Protein contact network where nodes or amino acids are depicted as red circles and edges or interactions as blue lines.

TABLE I: Linear controllability evaluation

	Multi State Proteins			Two State Proteins			Chromosome
	a6n	adw	aon	aps	arr	avz	
Driver nodes	9	7	18	5	3	5	138
size of the system	58	35	95	38	15	15	765
Normalized driver nodes size	0.1552	0.2	0.1895	0.1315	0.2	0.3334	0.1804

TABLE II: Non-linear (FVS-C) controllability evaluation

	Multi State Proteins			Two State Proteins			Chromosome
	a6n	adw	aon	aps	arr	avz	
Driver nodes	38	28	77	33	9	11	552
size of the system	58	35	95	38	15	15	765
Normalized driver nodes size	0.6551	0.875	0.8105	0.8684	0.6	0.7333	0.6823

TABLE III: Non-linear (MDS-C) controllability evaluation

	Multi State Proteins			Two State Proteins			Chromosome
	a6n	adw	aon	aps	arr	avz	
Driver nodes	2	2	1	1	5	1	8
size of the system	58	35	95	38	15	15	765
Normalized driver nodes size	0.0344	0.0571	0.0105	0.0263	0.3334	0.0666	0.0104

numbers for multi-state proteins and two-state proteins.

Also, in Table 4. we can see the nodes which appear to be driver nodes irrespective of the model/parameter.

To test the correlation between the number of driver nodes and the folding rate, we provided a table containing the correlation and a plot for each method and matrices (Contact maps and Topology matrices).

A. Contact maps

We test three analysis methods for driver nodes identifications as we mentioned above: 1. Linear controllability, 2. FVS (non-linear) and 3. Minimum dominating set (non-linear).

In 2/3 cases, we have a mild positive correlation between (normalized) driver nodes and folding rate and difference in

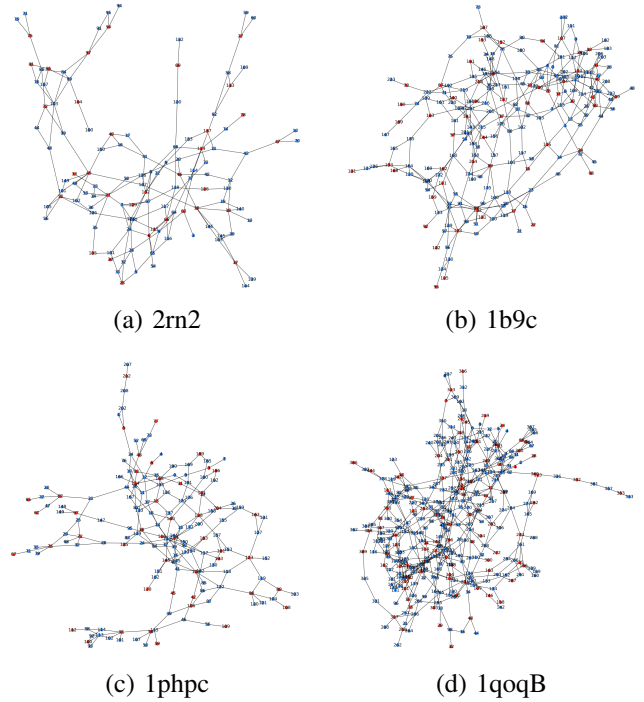


Fig. 13: Non-linear (MDS-C) controllability evaluation of: 2rn2, 1b9c, 1phpc and 1qoqB. The driver nodes are highlighted in red.

TABLE IV: Driver nodes irrespective of the model/parameter choice

Protein	driver nodes
a6n	13, 18, 25, 29, 48, 55.
adw	12, 15, 17, 25.
aon	2, 10, 14, 21, 25, 30, 40, 73, 74, 87.
aps	8, 31.
arr	8
Chromosome	15, 16, 28, 51, 63, 71, 89, 91, 103, 111, 118, 122, 135, 146, 149, 162, 171, 183, 184, 188, 189, 190, 199, 207, 226, 239, 241, 244, 250, 254, 256, 264, 271, 282, 331, 349, 350, 361, 364, 371, 378, 379, 385, 426, 430, 432, 436, 439, 445, 463, 464, 480, 486, 489, 504, 507, 538, 542, 559, 560, 569, 577, 597, 639, 647, 665, 670, 677, 678, 679, 680, 687, 708, 711, 714, 726, 765.

driver node distribution between two and multi-state folders.

1) *Linear controllability*: The distribution of the number of driver nodes for two protein folding types, multi-state and two-state, is displayed in Fig. (14).

Correlation between the number of driver nodes and folding rates are also displayed in Fig. (15) for both protein folding types.

2) *FVS*: The distribution of the number of driver nodes for two protein folding types, multi-state and two-state, is displayed in Fig. (16).

Correlation between the number of driver nodes and folding rates are also displayed in Fig. (17) for both protein

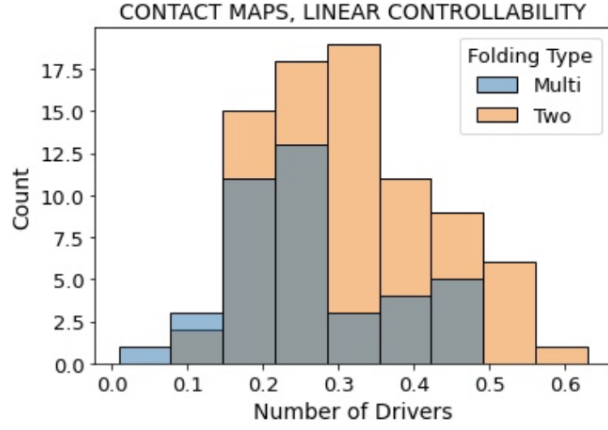


Fig. 14: The two distributions do differ, although not dramatically.

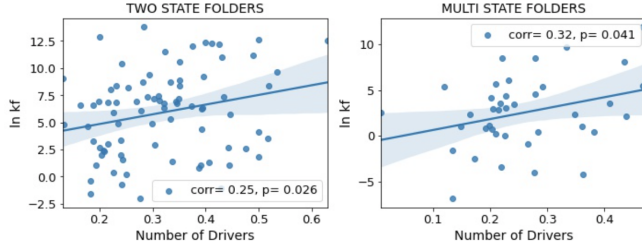


Fig. 15: we can see that there is mild correlation between $\ln k_f$ and number of drivers for the contact maps.

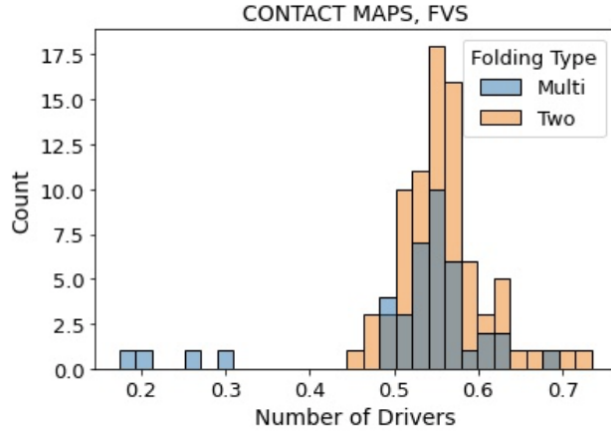


Fig. 16: No difference between the two distributions.

folding types.

3) *Minimum Dominating Set*: The distribution of the number of driver nodes for two protein folding types, multi-state and two-state, is displayed in Fig. (18).

Correlation between the number of driver nodes and folding rates are also displayed in Fig. (19) for both protein folding types.

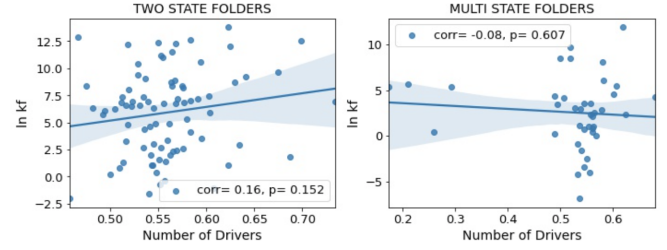


Fig. 17: we can see that there is no correlation between $\ln k_f$ and number of drivers for the contact maps.

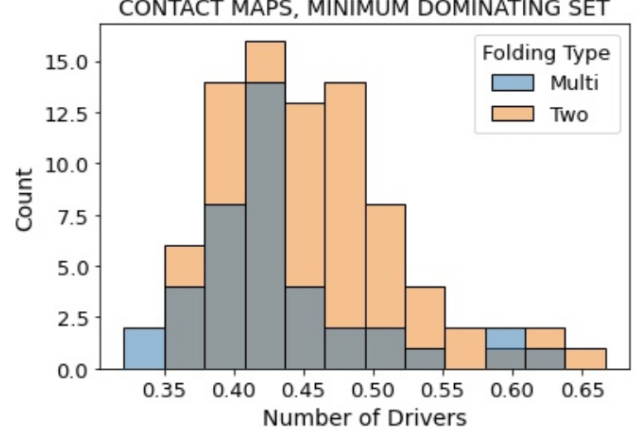


Fig. 18: The two distributions do differ, although not dramatically.

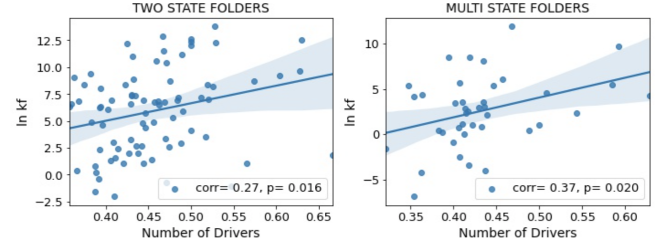


Fig. 19: we can see that there is mild correlation between $\ln k_f$ and number of drivers for the contact maps.

B. Topology matrix

We also test three analysis methods for driver nodes identifications as we mentioned above: 1. Linear controllability, 2. FVS (non-linear) and 3. Minimum dominating set (non-linear) for Topology matrices.

In 2/2 cases, we correlate (normalized) driver nodes and folding rate, although the two methods present ‘flipped’ plots, so we might want to look into what precisely these methods do.

Also, some outliers could spoil the correlation. Their correlations look more pronounced than with contact maps.

1) *FVS*: The distribution of the number of driver nodes for two protein folding types, multi-state and two-state, is displayed in Fig. (20).

There is a correlation between drivers and folding rate, but it looks like there might be some outliers to look into.

Since the relationship does not look very linear, we also calculated the Spearman correlation (Table. (V)).

TABLE V: Spearman correlation For Different protein folding type (Method: FVS)

	Multi state	Two state
Spearman correlation	-0.56	-0.51

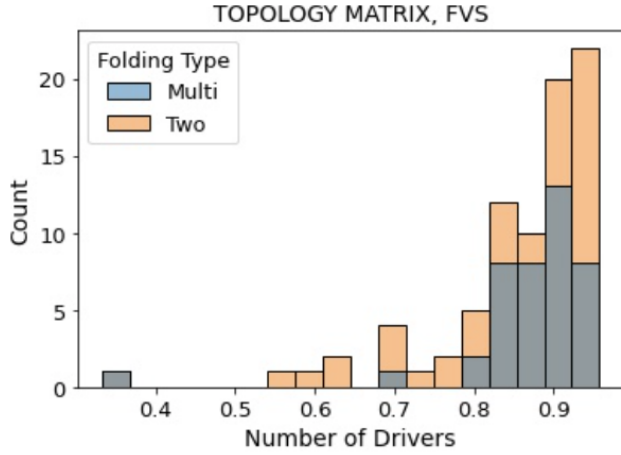


Fig. 20: No difference between the two distributions.

Correlation between the number of driver nodes and folding rates are also displayed in Fig. (21) for both protein folding types.

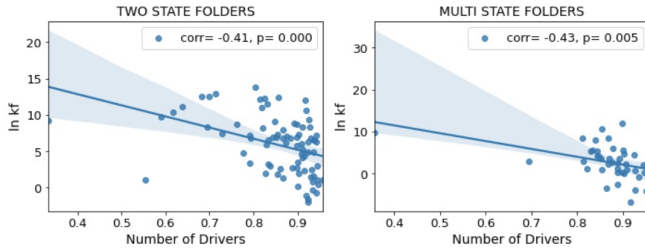


Fig. 21: we can see that here is correlation between drivers and folding rates for the Topology matrices.

2) *Minimum Dominating Set*: The distribution of the number of driver nodes for two protein folding types, multi-state and two-state, is displayed in Fig. (22).

There is a correlation between drivers and folding rate, but it looks like there might be some outliers to look into.

Since the relationship does not look very linear, we also calculated the Spearman correlation (Table. (VI)).

Correlation between the number of driver nodes and folding rates are also displayed in Fig. (23) for both protein folding types.

About the time complexity, we should consider that the Linear controllability, the feedback vertex set problem, and

TABLE VI: Spearman correlation For Different protein folding type (Method: MDS)

	Multi state	Two state
Spearman correlation	0.67	0.58

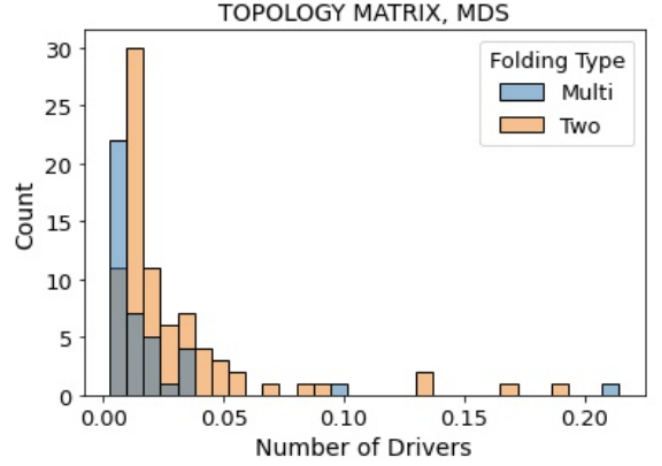


Fig. 22: The two distributions do differ.

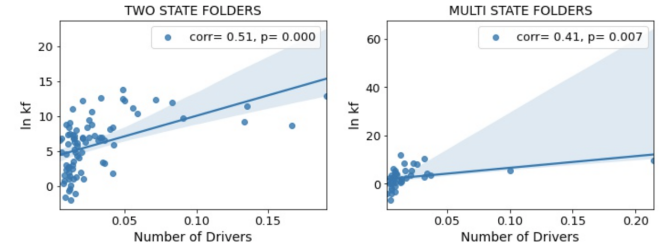


Fig. 23: we can see that there is correlation between drivers and folding rate, but it looks like there might be some outliers to look into.

the dominating set problem are NP-complete problems in computational complexity theory. For example time complexity for the Linear controllability is $O(N^2(\log(N)).2)$.

But the codes and algorithms which was written by the team to extract the driver nodes are very efficient and fast for instance we used integer linear programming (ILP) method which is a mathematical optimization program. Driver nodes extraction for the chromosome dataset would only take a few seconds to run for all three methods.

REFERENCES

- [1] Kalman, R. E. (1960). "A New Approach to Linear Filtering and Prediction Problems". Journal of Basic Engineering. **82**, 35-45.
- [2] Bahadorian, M., Alimohammadi, H., Mozaffari, T., Tabar, M., Peinke, J. and Lehnertz, K., 2019. A topology-dynamics-based control strategy for multi-dimensional complex networked dynamical systems. Scientific Reports, 9(1).
- [3] Gustav Lindmark, Controllability of Complex Networks at Minimum Cost, Linköping Studies in Science and Technology Dissertation, DOI: 10.3384/diss.diva-165258
- [4] Liu, Yang-Yu and Barabási, Albert-László, Control Principles of Complex Networks, DOI:10.1103/RevModPhys.88.035006 (2016).

- [5] Mochizuki, A., Fiedler, B., Kurosawa, G., & Saito, D. (2013). Dynamics and control at feedback vertex sets. II: A faithful monitor to determine the diversity of molecular activities in regulatory networks. *Journal of theoretical biology*, 335, 130-146.
- [6] Zhang, XF., Ou-Yang, L., Zhu, Y. et al. Determining minimum set of driver nodes in protein-protein interaction networks. *BMC Bioinformatics* 16, 146 (2015).
- [7] Jose C Nacher and Tatsuya Akutsu 2012 *New J. Phys.* 14 073005
- [8] Kalman, R. E. (1963), *J. Soc. Indus. and Appl. Math. Ser. A* 1 (2), 152.
- [9] Lin, C.-T. (1974), *IEEE Trans. Auto. Contr.* 19 (3), 201.
- [10] Chen, Y.-Z., L. Wang, W. Wang, and Y.-C. Lai (2015), [arXiv:1509.03196v1](https://arxiv.org/abs/1509.03196v1).
- [11] Yan, G., G. Tsekenis, B. Barzel, J.-J. Slotine, Y.-Y. Liu, and A.-L. Barabási, (2015), *Nat Phys* 11 (9), 779.
- [12] Chen Y-Z, Wang L-Z, Wang W-X, Lai Y-C. 2016 Energy scaling and reduction in controlling complex networks. *R. Soc. open sci.* 3: 160064. <http://dx.doi.org/10.1098/rsos.160064>
- [13] Chen YZ, Wang LZ, Wang WX, Lai YC. 2015 The paradox of controlling complex networks: control inputs versus energy requirement. (<http://arxiv.org/abs/1509.03196v1>)
- [14] Guoqi Li et al, Minimum-cost control of complex networks, doi:10.1088/1367-2630/18/1/013012
- [15] A.E. Isaac, A. Arumugam, Protein contact maps: A binary depiction of protein 3D structures, *Physica A* (2016), <https://doi.org/10.1016/j.physa.2016.08.033>.